

Pranav Raghavendra Rao

pranav.raghavrao@gmail.com | +1 (203)-673-1028 | [LinkedIn](#) | [GitHub](#)

EDUCATION

Northeastern University, Khoury College of Computer Sciences – M.S. in Data Science, 4.0 GPA Jan 2024 – Apr 2026
B.M.S. College of Engineering – B.E. in Electrical & Electronics Engineering, 4.0 GPA Aug 2018 – July 2022

SKILLS

Programming and Databases: Python, TypeScript, R, C, C++, MATLAB, MySQL, SQL Server, PostgreSQL, MongoDB
AI/ML Tools: NumPy, TensorFlow, PyTorch, CUDA, OpenAI API, WebSockets, LangChain, LangGraph, LangSmith, RAGAS
Generative AI & LLMs: Inference Optimization, Multimodal AI, RAG, Agentic Systems, Prompt Engineering, Vector Databases
Infrastructure: Spark, Azure (Databricks, ML Studio, ADLS), AWS (SageMaker, S3, ECS, Fargate, ElastiCache, RDS)
MLOps: Git, Docker, FastAPI, Terraform, GitHub Actions, MLflow

EXPERIENCE

AI Research Engineer - Associate, Yale University – New Haven, USA Aug 2026 – Present

- Building an **LLM structured generation pipeline** converting research grant narratives into IRB-compliant protocol documents, producing **Pydantic**-validated checklist items across regulatory requirements to drive workflow optimization for PIs
- Engineering a centralized **clinical data platform** on PostgreSQL, consolidating **multimodal research data** for cohort-level querying, supporting MNE-Python EEG preprocessing and predictive modeling for psychosis risk prediction

AI Engineer Co-op, Boehringer Ingelheim – Ridgefield, USA Aug 2025 – Dec 2025

- Built a **LangChain RAG agent** over Change Requests, CAPA reports, and SOP-linked records with Pinecone, **deployed** behind a containerized FastAPI service on **AWS** with CI/CD, cutting **QA mean time-to-resolution (MTTR)** from 3 days to under 1 day
- Pretrained a **LLaMA-based LLM** on 600M+ visits across 800K+ patients, outperforming traditional ML baselines on Interstitial Lung Disease (ILD) progression, distributed across an 8x GPU cluster with **PyTorch** DDP and PagedAttention-based **vLLM** inference
- Managed the training lifecycle across baseline and transformer variants on **MLflow**, enabling reproducible model selection

ML Engineer Intern, Berkshire Hathaway Energy (MidAmerican) – Des Moines, USA June 2025 – Aug 2025

- Developed and deployed a **production-grade LSTM-Transformer autoencoder** for wind turbine **anomaly detection** across 1B+ SCADA **time-series records**, contributing to a 12% fleet availability improvement across 3400+ wind assets
- Engineered scalable batch inference pipelines using **Azure Databricks** for **distributed processing** and **Azure ML managed endpoints** for fleet-wide model serving and automated scoring
- Built a **turbine health** scoring framework aggregating feature-level **reconstruction errors** into per-asset health scores, ranking 3,400+ wind assets by degradation severity for maintenance prioritization

Deloitte – Bengaluru, India Feb 2022 – Dec 2023

Data Scientist (Sep 2022 – Dec 2023)

- Optimized a **regression-based recommendation model** for retail assortment optimization, recommending Stock Keeping Units (SKUs) across retailer channels and achieving a 20% increase in revenue through improved inventory efficiency
- Developed a **K-Means segmentation** model to cluster retailer channels across 500+ cities by tier, enabling highly targeted **go-to-market strategies** across distinct channel profiles and regional market segments
- Conducted statistical **hypothesis testing** on ad-hoc growth initiatives, surfacing 3 high-potential market segments and driving targeted expansion decisions backed by statistically validated insights
- Led successful **migration** from **Pandas** to **PySpark** dataframes, boosting data processing efficiency and cutting client costs by 40%

Analyst Trainee (Feb 2022 – July 2022)

- Automated Qlik Sense to Snowflake SQL migration, streamlining analytics workflows and enabling scalable data integration
- Designed scalable ETL pipelines for entertainment data, reducing processing time by 30%, and analyzed trends using Power BI

AI Intern, CellStrat – Bengaluru, India Sept 2021 – Jan 2022

- AI product development for autonomous driving, leveraging lane and object detection models to enhance driving safety

PROJECTS

Real-Time Multimodal Voice Agent | *OpenAI Realtime API, WebSockets, Terraform, AWS* Apr 2026

- Deployed a real-time voice agent on ECS Fargate with Terraform, routing WebSocket audio through a FastAPI backend to the OpenAI Realtime API with Redis session state and PostgreSQL conversation persistence

Intent-to-Calendar Scheduling Agent | *LangGraph, Google Calendar API, FastAPI, Docker, GitHub Actions* Mar 2026

- Deployed a multi-user LangGraph scheduling agent on AWS ECS Fargate with Terraform and LangSmith agent tracing, converting natural-language input into calendar events through a FastAPI backend with per-user OAuth and async PostgreSQL

GitHub Repository Explainer - MCP Server | *Python, GitHub API, PyPI* Feb 2026

- Built and published a Python MCP server on PyPI, enabling any MCP-compatible AI coding environment to access live GitHub repositories via tools spanning repo summarization, file explanation, freeform Q&A, and dependency auditing